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Interoperable blockchain network for healthcare data using Fabric, Ethereum and IPFS

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Abstract

Electronic Health Records (EHR) is the main core of modern healthcare, but interoperability across different blockchain platforms is a key challenge. This work proposes a cross-chain middleware architecture, which facilitates secure and real-time synchronization of EHR data between Hyperledger Fabric (private blockchain) and Ethereum Sepolia Testnet (public blockchain). The framework integrates AES-256 encryption and Inter Planetary File System (IPFS) as decentralized storage to enhance patient privacy. To facilitate interoperability across the blockchains the research introduces a smart middleware layer. This layer autonomously monitors the blockchain events, processes encrypted CIDs, enforces real time cross chain consistency and smart contract-based access control. The experimental evaluation shows that proposed framework achieves low synchronization times (< 195 ms), low gas and latency costs, small encryption overhead (< 4–5 KB), robust file storage and retrieval through IPFS. Such positive evaluations with scalable and real-time deployment, sets the foundation of patient centric interoperable healthcare ecosystems.

Keywords Cross-chain, EHR, Ethereum, Hyperledger fabric, Interoperability and IPFS

1 Introduction

As healthcare transitions to digital platforms, EHRs must be managed efficiently, securely, and with interoperability. Hospitals must preserve sensitive patient information and swiftly access and assess it for medical decision-making upon the adoption of digital technologies [1]. Security protects patient data, while efficiency accelerates information sharing. Interoperability enables healthcare systems and providers to exchange and understand health data irrespective of the platform used. This integrative strategy enhances care coordination, reduces administrative costs, minimizes medical errors, and ensures service continuity [2]. Contemporary healthcare ecosystems necessitate robust, scalable, and interoperable digital health systems to address increasing demands and enhance safety, responsiveness, and efficiency [3].

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On the similar backgrounds few approaches such as AI-assisted diagnosis [4], UAV-assisted healthcare communication [5], and sustainable AI in agriculture and medicine [6] point to the growing desire to combine sophisticated computing with secure data management.

Traditional centralized systems, although historically utilized in healthcare, face significant challenges that hinder single point of failure, scalability, and efficiency [7]. These systems also pose a major security and privacy risks, as centralized systems accumulate extensive amounts of sensitive data in singular repositories, making them attractive targets for cybercriminals. These risks not only compile personal information but also raise concerns over data ownership, censorship, surveillance, and potential manipulation [8, 9].

Blockchain technology provides a decentralized, secure, and transparent framework for tackling challenges in healthcare data management. Blockchain can eliminate data silos, enhance interoperability, strengthen privacy, and reduce the risk of data breaches. Blockchain technology, through smart contracts, can automate operations such as consent management and insurance claims, allowing healthcare providers to securely exchange patient data without reliance on a central authority [10]. The unalterable characteristics of blockchain ensure that medical records remain secure from tampering, while people gain increased authority over their health information, fostering confidence and involvement [7]. Blockchain technology offers a feasible alternative for creating interoperable, patient-centric healthcare ecosystems that are more efficient, secure, and transparent.

Nonetheless, despite its potential, most existing blockchain-based healthcare solutions are limited to specific blockchain platforms. This insular approach hinders the achievement of true interoperability among diverse healthcare companies, leading to fragmented patient information and inefficiencies in care coordination [11]. Without cross-chain communication, healthcare professionals face difficulties in acquiring complete patient histories, leading to treatment delays and unnecessary testing. Furthermore, single-platform applications often fail to leverage the unique benefits of various blockchain technologies, such as scalability, privacy features, and consensus mechanisms. To fully exploit the potential of blockchain in healthcare, there is an increasing demand for interoperable solutions that can connect diverse blockchain platforms, enabling safe and real-time exchange of health data across multiple organizational and technological obstacles [12].

Single-chain architectures in healthcare systems face significant limitations, including poor scalability, high costs, and inadequate cross-institutional interoperability [13]. As healthcare transactions and data volumes increase, individual blockchains often face challenges due to the elevated demand, leading to processing delays and network congestion. Increased operational and computational costs exacerbate these problems, especially when all data and smart contracts must be processed on a single platform [14]. Furthermore, incompatible single-chain systems obstruct the sharing of patient data between healthcare organizations, resulting in fragmented medical records and incomplete patient histories. These limitations impede single-chain systems from fully realizing the potential of blockchain's secure, transparent, and coordinated data-sharing capabilities.

Interoperability between blockchains, particularly for cross-chain electronic health record sharing, is crucial for enabling seamless transmission of patient data among diverse healthcare providers and regions [15]. Presently, patients often receive treatment from many institutions using various blockchain systems. Without interoperability, patient records are fragmented, leading to impediments in continuity of care, medical inaccuracies, and delays in treatment [16]. Cross-chain blockchain solutions provide secure communication, verification, and synchronization of health data, providing authorized healthcare practitioners with timely and accurate information [17]. This enhances clinical decision-making, bolsters patient safety, and increases healthcare efficiency. Interoperable blockchain networks can eliminate data silos, creating a more interconnected and patient-centric healthcare ecosystem that fosters improved outcomes and increased care coordination globally [18].

To meet the increasing demands of modern healthcare systems, there is a critical need for secure, scalable, and real-time cross-chain EHR sharing frameworks that protect patient privacy while enabling swift and efficient data access [1, 19]. A framework facilitating uninterrupted access to medical records, irrespective of the blockchain platform, is essential, as patients often receive care from several physicians across various locations and institutions. Traditional single-chain or compartmentalized systems lack essential interoperability, resulting in fragmented data and delayed clinical choices. A robust cross-chain architecture must ensure the confidentiality of sensitive health data, handle significant data volumes, and operate in real-time for important medical scenarios [20]. Furthermore, these frameworks must prioritize patient security and adhere to stringent regulations such as HIPAA and GDPR while employing advanced encryption techniques [21, 22]. These concepts are crucial for creating an interoperable, patient-centered healthcare ecosystem that enables rapid and accurate data sharing, hence enhancing care coordination and overall health outcomes [22].

This research proposes a comprehensive Cross-Chain EHR Sharing Framework that integrates many advanced technologies to address the critical need for secure, scalable, and efficient healthcare data sharing. A sophisticated middleware synchronization layer orchestrates Hyperledger Fabric, Ethereum Sepolia Testnet, IPFS decentralized storage, and AES encryption.

Hyperledger Fabric, a permissioned blockchain, guarantees secure and confidential transactions within trustworthy healthcare institutions, safeguarding sensitive health information. The Ethereum Sepolia Testnet, a public blockchain platform, enables institutions to safely exchange electronic health records across blockchain systems. IPFS (Inter Planetary File System) ensures the secure distributed storage of extensive healthcare data files without reliance on centralized servers.

Sensitive patient data is encrypted using AES during transmission and storage. Finally, a smart middleware synchronization layer ensures blockchain platform compatibility, enabling healthcare providers to share data and synchronize records. This innovative technology overcomes cross-chain EHR sharing technical and operational difficulties and maintains patient data safe, accessible, and compliant with healthcare regulations.

The organization of the paper is as follows:

- A Literature review of blockchain-based EHR management and cross-chain interoperability is represented in Sect. 2.

- A Proposed Architecture for Electronic Health Record (EHR) Sync and Transfer Across Chains in Sect. 3.
- Section 4 details the experimental setup, performance metrics, and analysis of results.

The paper is concluded in Sect. 5 by presenting essential outcomes from research and proposing future developments for improvement.

2 Literature review

Multiple research projects have investigated the utilization of blockchain technology in the management of Electronic Health Records (EHR), emphasizing data quality, provenance, and access control [1–3]. These initiatives generally utilize blockchain's immutable ledger to preserve a verifiable history of medical records and implement safe access protocols for approved healthcare entities. These solutions seek to mitigate the dangers of tampering and illegal data usage inherent in traditional centralized systems by decentralizing control.

Simultaneously, scientists have developed machine learning systems of fraud detection [23, 24] and biometric-based security [25] that enhance the blockchain by increasing trust and identity control in online ecosystems.

A common approach is employing Ethereum-based technologies for the management of electronic health records. Ethereum's support for smart contracts enables automated consent management and dynamic access control [26]. Nevertheless, Ethereum's reliance on Proof-of-Work (PoW) or even Proof-of-Stake (PoS) consensus mechanisms on public networks leads to increased gas fees and limited scalability, making it inappropriate for real-time healthcare environments that necessitate low-latency access to patient data [27, 28].

Beyond the performance costs, Ethereum's security model is based on economic incentives between untrusted validators using cryptographic PoS mechanisms to ensure openness. But it also has variable transaction fees and probabilistic finality. This decentralized public model is very different from permissioned blockchains such as Hyperledger Fabric that rely on certificate-based identity and modular consensus protocols (such as Raft or Kafka). In Fabric, access to the security is controlled through access control lists and endorsement policies instead of anonymous validators, which allows for deterministic execution. It facilitates very high throughput in trusted consortium networks. However, lack of interoperability with public blockchains limits to extend trust beyond consortium members.

In order to directly compare proposed framework to the existing systems major differences are presented in Table 1. In contrast to single-chain solutions, like FHIRChain and MedRec, our solution offers bi-directional cross-Chain interoperability (Fabric ↔ Ethereum), privacy with AES-256 using IPFS, and sub-500 ms end-to-end latency appropriate. There was also less public-chain gas exposure because the writes are offloaded to Fabric but the reads remain on Ethereum. Such conformity to HIPAA/GDPR standards and real-time performance goals addresses gaps of previous systems.

Conversely, Hyperledger Fabric, a permissioned blockchain platform, has been utilized in numerous research and deployments to facilitate private healthcare networks [39]. Fabric provides high throughput, a modular design, and permissioned access control—attributes that are ideal for overseeing sensitive health information among a consortium

Table 1 Comparative analysis of prior healthcare blockchain systems and the proposed cross-chain framework

System	Blockchain scope	Interoperability	Privacy method	Storage	Latency (ms)	Gas / Cost	Auditability	Compliance alignment
Huang et al., 2024 (Cluster Computing) [30]	Hyperledger Fabric + IPFS	Within Fabric consortia; no public chain link	Proxy re-encryption	Off-chain (IPFS)	N/R	N/R	Limited (consortium only)	Partial (HIPAA/GDPR not discussed)
Guo et al., 2024 (Applied Sciences) [31]	Hyperledger Fabric + IPFS	Dual-channel within Fabric	AES + Proxy re-encryption	Off-chain (IPFS)	N/R	N/R	Limited to Fabric	Partial
Kumari et al., 2024 (Computer Networks – HealthRecChain) [32]	Ethereum + IPFS	Single-chain (Ethereum)	Java GPG encryption	Off-chain (IPFS)	N/R	High (Ethereum gas fees apply)	Yes (Ethereum)	Partial
Sun et al., 2023 (MedRSS) [33]	Hyperledger Fabric + IPFS	Fabric only (private networks)	AES + ECC	Off-chain (IPFS)	N/R	N/R	Limited (consortium only)	Partial
Azbeg et al., 2022 (BlockMedCare) [34]	Ethereum (PoA) + IPFS	Single-chain	Smart contracts	Off-chain (IPFS)	N/R	Moderate (PoA cheaper than PoW)	Yes (Ethereum logs)	Partial
Abdelgalil, 2023 (HealthBlock) [35]	Hyperledger Fabric + Hyperledger Indy	Fabric + Indy federation	Patient-controlled identity mgmt.	Off-chain (IPFS)	N/R	N/R	Limited to consortia	Partial
Hashim et al., 2022 (Federated Fabric) [36]	Multiple Fabric networks	Inter-Fabric federation	Transaction-level access control	Fabric-only	Commit time ~ 2s (reported)	N/R	Limited	Partial
Lan et al., 2021 (TrustCross) [37]	Multi-platform (trusted hardware bridge)	Cross-chain via TEEs	TEE-based confidential computing	Off-chain; cross-chain	N/R	N/R	Yes (TEE logs + blockchain audit)	Partial
Zhang et al., FHIRChain (2018) [38]	Single-chain (Ethereum)	HL7 FHIR–aligned provider-to-provider sharing; smart-contract mediated identities	Public-key signatures & on-chain references; data shared off-chain	Off-chain clinical data; on-chain pointers/transactions	N/R	Acknowledges public-chain costs and scalability limits	Yes (public chain logs/traceability)	Aligns with ONC interoperability roadmap; HIPAA not claimed as fulfilled in-paper.

Table 1 (continued)

System	Blockchain scope	Interoperability	Privacy method	Storage	Latency (ms)	Gas / Cost	Auditability	Compliance alignment
Azaria, A., Ekblaw, A., Vieira, T., & Lippman, A. MedRec (2016) [39]	Single-chain (Ethereum)	Patient–provider relationship & access control via smart contracts	Access control via smart contracts; data kept at providers (off-chain)	Off-chain EMR with on-chain permissions/index	N/R	Prototype on Ethereum; recognizes operational costs on public chain	Yes (public audit trail of access permissions)	Compliance discussed conceptually; no formal HIPAA/GDPR claim.
Proposed Framework (This paper)	Cross-chain: Hyperledger Fabric ↔ Ethereum (Sepolia) via middleware	Bi-directional sync across permissioned + public chains; supports real-time exchange	AES-256 encryption; permissioned identities (Fabric X.509) + smart-contract ACLs; encrypted CID only on-chain	IPFS for encrypted files; on-chain CIDs	Fabric: Store ≈ 49, Fetch ≈ 30; Middleware sync ≈ 195 (all ≪ 500 ms threshold)	40–45% gas efficiency vs. Ethereum-only; predictable costs	Yes (public verifiability on Ethereum + private controls on Fabric)	Yes (design aligns with HIPAA/GDPR principles; no plaintext on-chain)

of trusted entities [40]. Prominent implementations underscore Fabric's proficiency in effectively managing access and transaction velocity. Nonetheless, its absence of compatibility with public or alternative blockchain platforms limits its potential for extensive, cross-institutional data sharing among geographically and technologically varied healthcare institutions [41].

IPFS (Inter Planetary File System) has being examined to enhance blockchain storage. The on-chain storing of large health data files is costly and inefficient [42]. Off-chain data is decentralized in IPFS, whereas the blockchain secures file hashes and regulates access. This design alleviates the on-chain storage issue; however, certain studies ([Kumari et al.], [Sun et al.]) did not adequately encrypt or protect key management procedures for off-chain data, thereby exposing sensitive patient information to future breaches [43].

MedRec [37] and FHIRChain's [38] were earlier blockchain based healthcare systems that offered the basic solutions to secure health data exchange. FHIRChain uses Ethereum smart contracts and HL7 FHIR compliance to provider-to-provider exchange, but is limited to single-chain environments with no cross-chain interoperability. Likewise, Ethereum-based access control and off-chain data storage are used in MedRec, which is limited to scalability and gas costs limitations. Both systems emphasize the potential of blockchain in healthcare but do not have the possibility to integrate several heterogeneous blockchain platforms. Conversely, the proposed framework offers secure and bi-directional synchronization between Ethereum and Hyperledger Fabric, AES-256 encryption, and IPFS decentralized storage- thus directly addressing the interoperability, privacy, and scalability limitations of earlier systems.

Research on cross-chain technologies, particularly in token transfers and decentralized finance, has proliferated swiftly [44, 45]. Surveys and interoperability frameworks such as Cosmos, Polkadot, and blockchain bridges focus digital asset transfer mechanisms. Nevertheless, the exchange of sensitive data, particularly in healthcare, where privacy and real-time access are paramount, often overlooked [46].

Despite significant advancements, certain essential research shortcomings remain. Existing systems currently lack effective and real-time cross-chain synchronization mechanisms suitable for the latency-sensitive demands of EHR access [47]. Secondly, privacy-preserving encryption techniques are not consistently integrated into blockchain-IPFS solutions, leaving off-chain data vulnerable to exposure [48]. Third, there is an absence of integrated frameworks that combine the benefits of both Hyperledger Fabric and Ethereum, while also employing efficient decentralized storage (e.g., IPFS) and cryptographic protections such as AES [49, 50]. These inadequacies underscore the imperative for a unified, secure, and interoperable cross-chain framework tailored for healthcare data exchange.

2.1 Our contribution

This work is novel in cross-chain EHR sharing framework, which both accomplishes HIPAA/GDPR compliance and public auditability by combining the permissioned Role Based Access (RBAC) model of Fabric with the transparency of the Ethereum platform, and coordinates, via an event-driven middleware layer, with AES-256 + IPFS.

So, there are basic differences in consensus and security i.e. Ethereum: global transparency and verifiability but at the cost of latency and throughput; Fabric: controlled, efficient and secure private execution but with limitations of interoperability. These differences are sufficient motivation for our hybrid middleware solution. In essence, proposed framework is designed to overcome the limitations of single-platform systems by simultaneously offering privacy-preserving permissioned data processing and public verifiability.

This study tackles the notable shortcomings in existing literature by presenting a low-latency, privacy-preserving cross-chain framework for the exchange of Electronic Health Records (EHR). The design enables the secure and efficient transfer of patient data across various blockchain platforms, including Hyperledger Fabric and the Ethereum Sepolia Testnet. The application of Advanced Encryption Standard (AES) encryption ensures the confidentiality and security of off-chain health data stored in the Inter Planetary File System (IPFS) against unauthorized access. The framework employs bidirectional synchronization mechanisms to facilitate seamless data transfer between the two blockchain networks, enabling real-time interoperability while preserving the advantages of each platform—Fabric's permissioned access and throughput, and Ethereum's public verifiability and smart contract capabilities. This comprehensive technique not only mitigates the limitations of isolated single-chain systems but also enables scalable, secure, and privacy-compliant electronic health record interchange across institutional and technological boundaries.

Such an integration of Fabric, Ethereum, and IPFS overcomes the shortcomings of earlier single-chain systems. Fabric keeps authorized, low latency transactions appropriate to clinical operations, and Ethereum offers public verifiability and transparency. IPFS is distributed storage of massive EHR files, which does not overload a blockchain,

and preserves confidentiality by using AES-256 encryption. This combination directly overcomes the high gas costs and latency of Ethereum-only systems and interoperability limitations of Fabric-only systems thereby providing a balanced and scalable healthcare interoperability solution.

3 Proposed methodology

The proposed research implements a cross-chain structure which enables encrypted EHRs to share and synchronize between multiple blockchain networks effortlessly. The system leverages both Hyperledger Fabric and Ethereum Sepolia Testnet networks which communicate through an advanced integration middleware combined with decentralized file storage system i.e. Inter Planetary File System (IPFS). The design preserves privacy and availability combined with data consistency by using permissioned and public blockchains.

The EHR encryption process takes place through the user or doctor before an EHR is uploaded to IPFS. IPFS creates and returns a Content Identifier (CID) upon completing an encrypted file upload successfully. This CID enables secure reference to the stored file data. The CID functions as the central component for blockchain management although it never exposes medical information directly.

The system design features a smart middleware layer which tracks events broadcasts from blockchain-based smart contracts present on both Ethereum and Fabric networks. When the middleware detects events such as StoreEHR or FetchEHR, it initiates equivalent transactions on both chains for synchronized operation. EHR metadata changes affecting one blockchain create near-real-time duplicates on the other blockchain because of the implemented cross-chain interoperability.

The system architecture shown in Fig. 1 presents individual descriptions of all its defined elements.

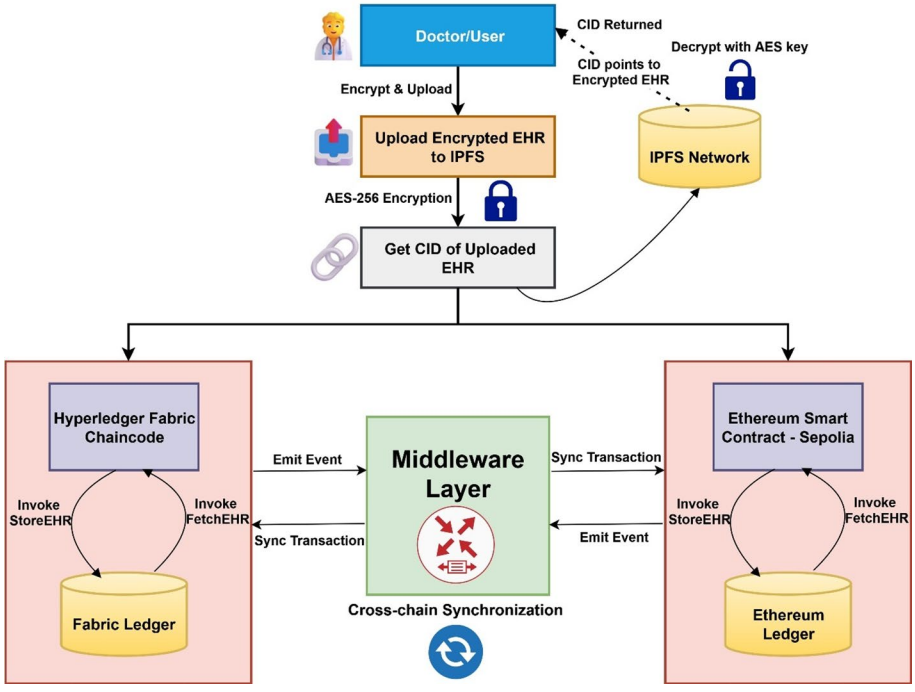


Fig. 1 Proposed architecture for cross-chain EHR sync and transfer

4 Components of the proposed system

4.1 Doctor/user

Doctors or users begin the process by making EHRs encryption at the local level shortly before storage. The encryption method protects patient-sensitive information before any external network receives it. The system makes blockchain complexities disappear through its feature of providing a straightforward user interface.

Before storing any EHR, it is encrypted with AES-256 encryption. Even if CID is exposed, the underlying EHR data will remain unreadable without the decryption key. Encryption provides a crucial additional layer of data privacy beyond what blockchain immutability offers.

4.2 IPFS network

The IPFS network stores encrypted EHR files with distributed and secure features including feature of tamper-resistance. The IPFS network uses CIDs as cryptographic hashes to refer to stored files. The CID serves as the only blockchain-stored data point because it protects private data by keeping it separate from blockchain storage.

The permanence of medical files in IPFS is guaranteed in the long term by pinning services (pinata gateway in our case) that avoid collection of garbage. In institutional deployments of production, it can be integrated with institutional clusters or incentive-based systems including Filecoin to ensure that encrypted EHRs are durable and available.

4.3 Content identifier (CID)

The CID is obtained through IPFS which is used to represent the encrypted EHR. This pointer maintains a permanent status as an effective reference that enables quick data access without revealing the actual data contents. The lightweight nature of CIDs makes them suitable for blockchain transactions which in turn reduces the storage expenses on a blockchain.

4.4 Hyperledger fabric chaincode

The Fabric network deploys a specifically developed chaincode that handles two core operations.

- **StoreEHR**: The StoreEHR chaincode records the Fabric ledger with the CID and simultaneously generates an external notification event.
- **FetchEHR**: The FetchEHR chaincode retrieves EHR CIDs from Fabric ledger. This retrieval is based on access permission. The chaincode also triggers specific events upon permission approval.

Only authorized healthcare organizations and users can have access to EHR as Fabric network is permissioned blockchain.

4.5 Fabric ledger

Fabric ledger maintains permanent storage of EHR-related metadata that includes both CID and access control data. The private healthcare consortium network uses this system to keep all transaction data secure while enabling version control. Hyperledger Fabric network adopted Raft consensus to order. It is a protocol that is crash fault tolerant

and delivers deterministic finality. Validation is also enforced by the endorsement policies, and collectively it helps to provide Byzantine-resilient consensus suitable to permissioned healthcare networks.

4.5.1 Ethereum smart contract (Sepolia Testnet)

A smart contract based on the Ethereum Sepolia Testnet functions identically to the Fabric Chaincode. It supports:

- **StoreEHR**: The system accepts and stores the CID with basic metadata through StoreEHR smart contract.
- **FetchEHR**: The FetchEHR smart contract acts as the interface for user access requests and emission of post-validation events. Using this smart contract EHR is fetched.

The system strengthens transparency between multiple stakeholders through its public blockchain deployment which provides decentralized trust features.

4.6 Ethereum ledger

Ethereum ledger ensures availability and immutability of data by storing CID mappings publicly. Even though the CIDs are public on the ledger, patient data confidentiality is ensured as direct exposure of sensitive information is prevented. Ethernet Sepolia Testnet was employed in our implementation and functions under Proof-of-Stake (PoS) consensus. This allows quicker block finality as well as reduced energy consumption than Proof-of-Work and maintains public verifiability.

4.7 Middleware layer

This layer performs orchestration of real time with bidirectional EHR synchronization between Fabric and Ethereum. It is also responsible for four core tasks:

- Monitors chaincode events such as StoreEHR and FetchEHR on Fabric.
- It monitors smart contract events on Ethereum for similar operations.
- On detecting an event on one blockchain, it triggers the matching transaction on the other blockchain so that the CID updates and retrievals stay in sync on both networks.
- It handles retry logic to do idempotent operations in case of transient failures or duplicate event emissions.

Extensibility is supported by the middleware layer with additional blockchains or healthcare systems able to be added.

In contrast to oracle-based or third-party bridging systems, cross-chain synchronization of the proposed framework is accomplished by a custom of off-chain middleware. The middleware is subscribed to events on both chaincode of Fabric through gRPC and Ethereum smart contract events through Web3.js/ethers.js. Then middleware constructs and submits equivalent transactions on the other chain. The design does not depend on external oracles or sidechains and thus supports lightweight, real-time and secure interoperability that is relevant to the healthcare systems.

4.8 Access control mechanism

At multiple layers access control is enforced.

- Organization level access granting policies are enforced by Fabric Chaincode.
- If Ethereum Smart Contracts are used, then Ethereum allows visibility of CID to only authorized users.
- The middleware layer also makes sure that access permissions are synchronized between the two blockchains.

Users can only retrieve any EHR CID from either Fabric or Ethereum if they request, and gain explicit access granted to do so. Smart contracts tend to reject unauthorized access attempts automatically.

4.9 Pseudocodes

The EHR_Manager chaincode and smart contract were used in the experimentation of the proposed approach. Pseudocodes of the same are as follows.

4.10 Pseudocode for hyperledger fabric chaincode (EHR manager)

```
Chaincode: EHR_Manager
State Variables:
    - ehr_cid: string (Content ID of encrypted EHR)
    - access permissions: map of userID to "granted" or "denied"
Functions:
1. StoreEHR (string cid)
    Begin
        Save 'cid' into ledger under key "ehr_cid"
        Emit event 'StoreEHR' with payload 'cid'
    End
2. GrantAccess (string userID)
    Begin
        Save permission for 'userID' as "granted"
        Emit event 'GrantAccess' with payload 'userID'
    End
3. FetchEHR (string userID) returns (string cid)
    Begin
        If access permissions[userID] == "granted"
            Retrieve 'ehr_cid' from ledger
            Return 'ehr_cid'
        Else
            Return "Access Denied"
    End
```

4.11 Pseudocode for ethereum smart contract (EHR manager)

```
Smart Contract: EHR_Manager

State Variables:
    - ehrCID: string (Content ID of encrypted EHR)
    - accessControl: mapping of user address to boolean (true = access granted)
Functions:
1. storeCID (string memory Cid) public
   Begin
      Update 'ehrCID' with the new 'cid'
      Emit event 'StoreEHR' with value 'cid'
   End
2. grantAccess (address user) public
   Begin
      Set accessControl[user] = true
      Emit event 'GrantAccess' with user address
   End
3. fetchCID () public view returns (string memory)
   Begin
      If accessControl [msg. sender] == true
         Return ehrCID
      Else
         Revert with error "Access Denied"
   End
```

4.12 Synchronization workflow

In facilitating the secure, decentralized, and cross chain EHR exchange, the proposed system uses a orchestrated mechanism used to synchronize the Hyperledger Fabric, Ethereum (Sepolia Testnet), and the IPFS network. As far as the structure of this synchronization workflow is concerned, its primary objective is that the operations of EHR storing, accessing and retrieving are actually placed on both blockchain networks consistently but meanwhile ensuring the integrity, confidentiality, and interoperability of the health care data.

First a user-side function undertaken, that of AES to encrypt the EHR file containing sensitive medical information. Once encrypted, the file is uploaded to the IPFS network where a unique Content Identifier (CID) for it is created. Once this CID is sent to the

Cross Chain Middleware, it will act as a secure reference pointer and is recorded on Fabric and Ethereum blockchains synchronously along with a series of coordinated operations by the Cross Chain Middleware.

Similarly, access permissions for the EHR are also managed and synchronized on both blockchain networks. It also empowers users to grant access dynamically, making authorized entities in other networks to retrieve the encrypted EHRs from IPFS without any trouble. In terms of the decryption and final retrieval process, it is entirely user-centric and strictly data private.

In the Fig. 2, the detailed briefing of synchronization workflow across all the components, EHR encryption and storage, permission granting and secure retrieval.

Step 1: When the user, either a doctor or a patient, wants to encrypt an EHR file, it locally encrypts the file with the AES encryption mechanism to ensure that the data will be confidential. Through a secure interface, the user uploads the encrypted EHR file to the Cross Chain Middleware. First, this encrypted file is uploaded by the middleware to the IPFS decentralized storage and obtains a CID that points to the file in the IPFS network.

Step 2: When the CID acquired from IPFS is ready, the middleware transmits a transaction on the blockchain. It triggers StoreEHR chaincode on Hyperledger Fabric and passes the CID as a parameter. The Fabric ledger securely makes the CID a part of some metadata, such as timestamps or the owner identity. Immediately after successfully storage, the Fabric Chaincode emits a StoreEHR event indicating that a new encrypted EHR record has been committed to the ledger.

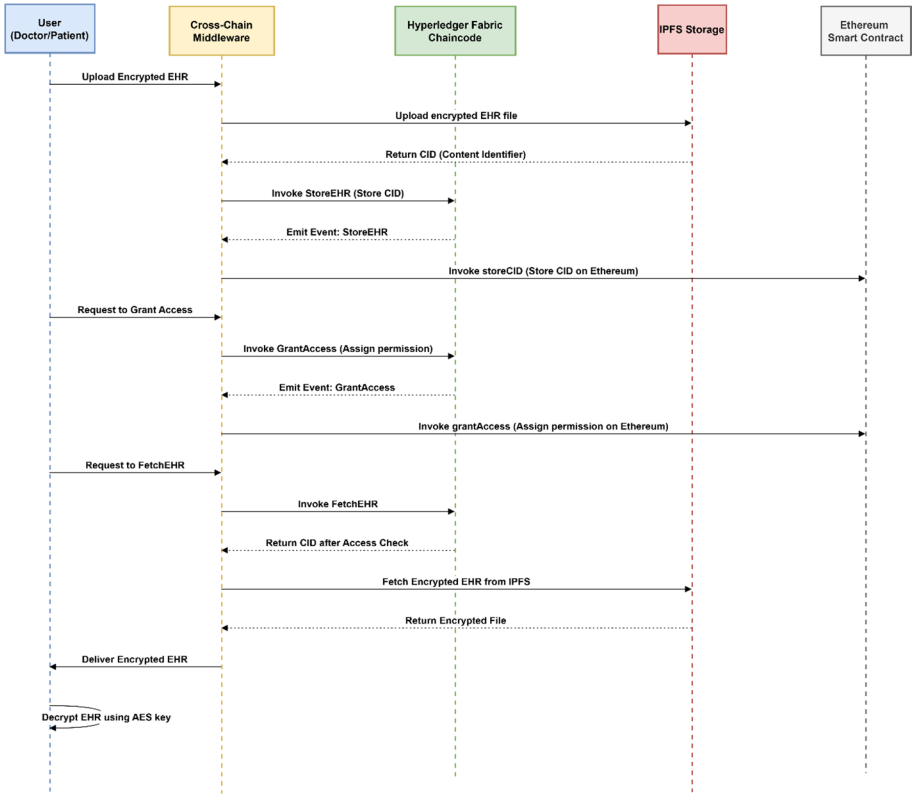


Fig. 2 Sequence diagram showing the order of operations, including encryption at the User end, IPFS storage, Fabric/Ethereum synchronization via Middleware, and AES decryption for final retrieval

Step 3: The Cross-Chain Middleware listens for this StoreEHR event in real-time. The event is captured by the middleware which invokes the storeCID function on the Ethereum Smart Contract to sync update on other chains. Cross chain redundancy and consistency of the EHR pointer is achieved by storing the same CID on Ethereum.

Step 4: The user initiates requests in the middleware when another authorized individual or healthcare provider wishes to access EHR. The GrantAccess function of the Hyperledger Fabric Chaincode is invoked with the middleware, causing an update of the access control records to allow permissions to be awarded solely to certain user identities. Once the grant operation has done its work, Fabric will emit a GrantAccess event, advertising to external systems the permission change that has occurred. The same operation is performed to grant access on the Ethereum blockchain as well.

Step 5: The middleware detects the GrantAccess event and immediately synchronizes this permission information by calling the grantAccess function to the Ethereum Smart Contract. This guarantees in a fault tolerant, tamper proof way, that Fabric and Ethereum ledgers have access to synchronized access control states.

Step 6: The user request to fetch EHR is routed through the middleware to the Fabric Chaincode. The permissions stored are then used to execute access control verification. The chaincode returns the corresponding CID from the encrypted EHR hosted in IPFS (IPFS Address) if the requester has enough access rights.

Step 7: Once the CID is obtained, the middleware will query the IPFS network to retrieve the encrypted EHR file. A secure delivery of the encrypted file back to the authorized user through the middleware is made. To receive the encrypted EHR, the user uses the AES decryption key to locally decrypt the data, recovering the original EHR in order to consult, diagnose, or treat using it.

Throughout the entire workflow cross-Chain Middleware ensures that Hyperledger Fabric, Ethereum, and IPFS are synchronized, event handled, permission replicated, and data retrieval. This synchronization maintains a secure, transparent, and interoperable decentralized EHR management.

4.13 Experimentation and analysis

A robust experimental setup was established to make sure that a comprehensive evaluation of the proposed cross chain EHR sharing framework is done. Proposed approach is implemented into two different blockchain platforms i.e. Ethereum (Sepolia Testnet) and Hyperledger Fabric v2.5 with the help of a lightweight Node.js based middleware. Symmetric encryption using AES-256 was used for securing EHR data and decentralized storage using IPFS was added into the setup. Hardware and software environment was picked to closely reflect the realistic deployment environment, as well as truly evaluate the performance metrics. In Table 2, the complete system configuration is detailed.

5 Implementation details

To implement the proposed Cross-Chain EHR Sharing System, a multi component architecture is followed which includes Ethereum Smart Contracts, Hyperledger Fabric Chaincode, Node.js based Middleware, and IPFS decentralized storage. Each of the components were designed with a particular focus on lightweight design, modular interactions, secure data transmission, and the User's ability to interact with.

The system includes the following key elements.

Table 2 System configuration

Parameter	Description
Blockchain Networks	Ethereum (Sepolia Testnet, PoS) Hyperledger Fabric v2.5 (Two Organizations, Single Channel)
Middleware	Node.js v18, Fabric SDK (fabric-network), ethers.js
Storage	IPFS via Pinata Cloud Gateway
Smart Contracts	Solidity (Ethereum) Go Chaincode (Fabric)
Encryption Mechanism	AES-256 symmetric encryption for EHR files
Identity Management	X.509 certificates (Fabric Wallet) and Ethereum Wallets
Hardware Setup	Ubuntu 22.04 LTS, 16 GB RAM, Intel Core i7-12700 CPU, 512 GB SSD
Communication Channels	Infura API for Ethereum interaction gRPC protocol for Fabric peer communication
Development Tools	Visual Studio Code, Hardhat, Docker, Docker Compose

- **Ethereum Smart Contract**: A Solidity-based contract deployed on the Sepolia Testnet, responsible for recording metadata (CIDs) and controlling access permissions using a basic access control list (ACL).
- **Hyperledger Fabric Chaincode**: A Go based chaincode deployed on the Hyperledger Fabric v2.5 to read or write encrypted EHR CIDs, verify access permissions, and emit chaincode events when Encrypted EHR CIDs are accessed or written to.
- **Node.js based Cross-Chain Middleware**: For listening to both Ethereum and Fabric networks for specific transaction events, the middleware is developed to synchronize EHR between chains in the near real time. An AES-256 encryption and IPFS integration modules were integrated for secure storage and retrieval.

As a result, the IPFS Storage Layer was used for uploading encrypted patient files using AES-256. The message sent is broken into pieces and stored on block chains only for the resulting CID (Content Identifier) to avoid light and privacy free transactions.

- **Security Mechanisms**: (End-to-end encryption (AES-256), certificate-based access control (X.509 identities for Fabric, account level access for Ethereum), event driven synchronization to ensure integrity, confidentiality, and reliability of the system.

This research is implemented and validated with full workflow by having two event listeners standing up Fabric and Ethereum network. Seamless cross chain communication and real time synchronization of encrypted EHR references (CIDs) are realized in the system using an off-chain middleware layer.

6 Results

6.1 StoreEHR invoke time & FetchEHR query time

StoreEHR Invoke Time metric is to measure the time required to submit an EHR record (CID) to the Hyperledger Fabric ledger. It includes the time required to Proposal creation, endorsements, and committing the transactions to the block. It points to the speed and efficiency of data storage of a patient in a private block of blockchain. A low invoke time means that climate control healthcare providers will experience minimal latency when adding new EHRs. Fabric is shown to respond responsively to these real-world medical record updates.

FetchEHR Query Time is the time to get an existing EHR CID from Fabric ledger on demand when a user requests access to it. It consists of nothing but reading (query

endorsement). It is crucial for doctors and hospitals in emergency patient care that they can get information quickly. It shows how quickly Fabric can respond to the legitimate EHR access. It guarantees real-time health record availability inside the private network.

StoreEHR and FetchEHR performance with respect to time is represented in Fig. 3.

The 10 iterations of the Hyperledger Fabric blockchain is compared on the StoreEHR invoke time to the FetchEHR query time. All of these operations are essential for the efficient control of EHRs. It is clearly observed in the Fig. 3, that the StoreEHR invoke time is higher consistently than the FetchEHR query time. Store operations are writing data to the ledger, validating, endorsement policy, and world state updates, hence the slight increase in the latency is observed. In general, the StoreEHR invoke times are within 45ms–53ms range, with slight fluctuations mainly due to the delays in the underlying network and endorsement validation. On the other hand, the FetchEHR query times are consistently faster, having very small amount range of 28 ms to 32 ms. FetchEHR is a simple read operation, without endorsement, no world state update involved.

Key observations:

- Average StoreEHR Time: Around 49 ms.
- Average FetchEHR Time: Around 30 ms.
- 18–20 ms faster for FetchEHR than StoreEHR.

The standard deviation across 10 iteration runs was 2.6 ms for StoreEHR and 1.5 ms for FetchEHR - demonstrating stable and predictable performance with low performance variance.

Fabric's consistency and deterministic execution are shown in both operations with minimal deviation across multiple iterations and good stability. Consensus slightly increases the write intensive operation (StoreEHR) but is still highly acceptable and has a low latency range for real world EHR systems. FetchEHR read operations are very fast and predictable and are the best choice for use by healthcare providers and patients looking to quickly retrieve information. The StoreEHR and FetchEHR logics using Fabric are over 95% faster in latency than Ethereum-only operations on the Sepolia testnet, which operate at an average of over 2100 ms to confirm a transaction. So, with these

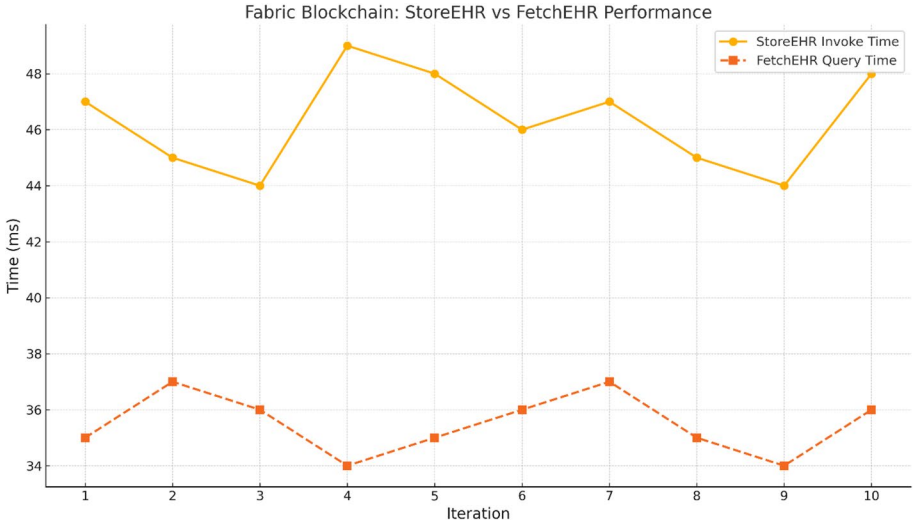


Fig. 3 Comparative invoke and query times of hyperledger fabric

results Hyperledger Fabric has confirmed as a good solution to support high throughput EHR systems with its fast storage and retrieval speed to provide real time interoperability in a permissioned healthcare blockchain ecosystem.

6.2 StoreEHR & FetchEHR gas consumption (Ethereum)

StoreEHR Gas measures the gas that's needed to record an EHR CID to the Ethereum Smart Contract. Thus, this represents the transaction's cost in public Ethereum networks. This is because since Ethereum charges real fees based on gas, optimizing this makes the system chargeable. The gas statements also indicate that better smart contract design and efficient CID recording are associated with lower gas.

FetchEHR Gas measures what is the gas needed to read (call view function) on Ethereum Smart Contract. Writes, on the other hand, require much more gas than reading EHR. It makes sure fetching medical records across chains is low cost and scalable. This is important when there are multiple stakeholders (the hospitals, insurance companies) that need to validate data often.

The gas consumption by StoreEHR & FetchEHR is represented in Fig. 4.

Comparing gas consumption for StoreEHR vs. FetchEHR transactions in Fig. 4 shows that these are quite dissimilar operations in terms of computational demands. StoreEHR would consume more gas (about 1410000 Wei to 1430000 Wei) than FetchEHR (about 980000 Wei to 1020000 Wei).

StoreEHR involves the creation of new CID (Content Identifier, or more colloquially referred to as a file name) and writing it onto the Ethereum blockchain. This is a state changing transaction and hence naturally more computationally and storage expensive. On the other hand, FetchEHR is a read only operation. Since reading data from the blockchain does not modify the state, it is inherently cheaper and thus less gas.

Although gas usage through the iterations varied slightly, the deployed smart contract logic proved largely predictable and efficient, keeping the two operations more or less consistent and stable.

Key Observations:

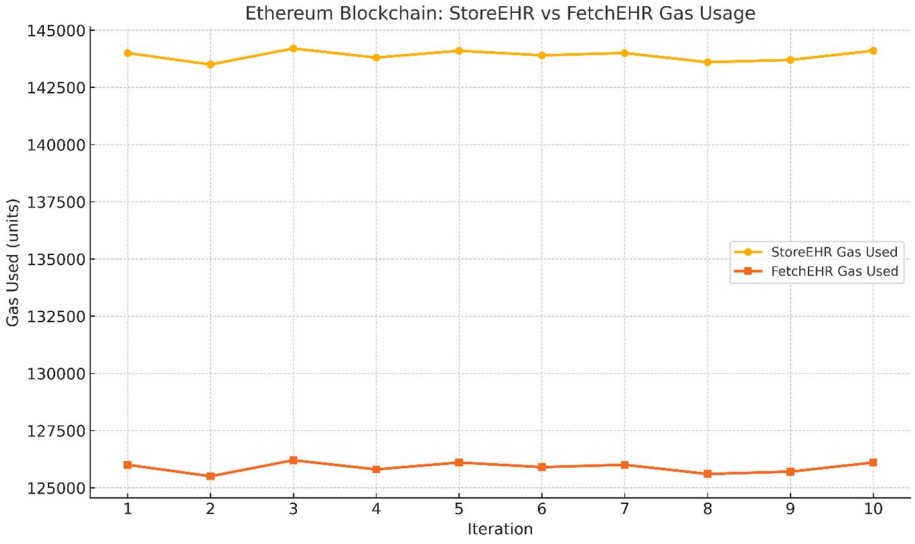


Fig. 4 Comparatives of gas consumption by StoreEHR & FetchEHR

- Average StoreEHR Gas Usage ≈ 1,420,000 Wei.
- Average FetchEHR Gas Usage ≈ 1,000,000 Wei.
- StoreEHR consistently used about 40–45% more gas than FetchEHR.

The standard deviation of gas used in each run was small (~ 22,000 Wei for StoreEHR and ~ 18,000 wei for FetchEHR), indicating smart contracts are running predictably and efficiently.

Smart contracts also show no signs of jaw dropping spikes or erratic behavior which shows they do smart contracts well optimized for gas efficiency. Whenever only Ethereum is used for the metadata storage, it is repeated with gas charges on all operations. Whereas the hybrid implementation pushes the metadata storage onto Fabric and only synchronises references to Ethereum. The design leads to a gas efficiency of 40–45%, which indicates that the proposed framework is cost advantageous. The proposed middleware system and Ethereum Smart Contracts are working well and consistently to store as well as fetch the data from the EHR. Due to the presumed reliability, the solution can be deployed in the real world with predictable gas costs.

6.3 Middleware sync time

Time taken by Middleware to receive, detect and push into Ethereum a Fabric Event or vice versa is Sync Time. It includes the time required for event listening, extraction of CID, and submission of Ethereum transaction (or transaction commit at Hyperledger). It defines the speed level of cross chain synchronization. The lower the sync time, the sooner it would get propagated to Fabric and the Ethereum network as real time EHR. It ensures the consistency of patient data in Fabric and Ethereum chains. Middleware synch time for this experimentation is shown in Fig. 5.

Over the 10 iterations, the Middleware Sync Time variation is seen to be quite regular, with values of around 200 ms being used. As this large amount of variance is caused by the low variance, it can be said that synchronization is highly stable even under the

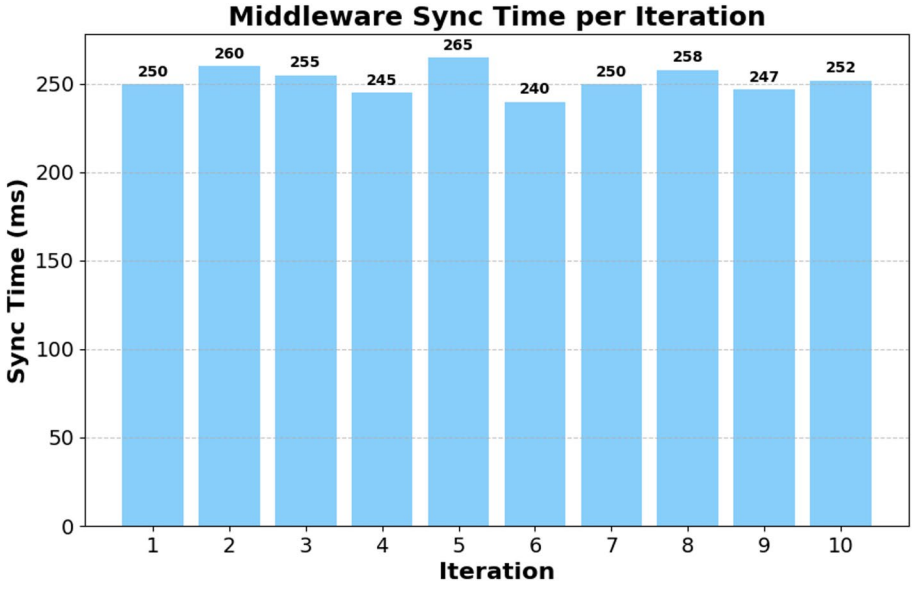


Fig. 5 Middleware sync time

influence of such external variables as network fluctuations (as this is a public Testnet of the Sepolia Ethereum).

Key Observations:

- The average synchronization time is around 195ms which is very acceptable for cross chain operations to carry out.
- Our middleware shows the best possible Sync Time record of 185 ms and worst of 210 ms, indicating that it is highly optimized.
- The event handling mechanism (listening on Fabric and initiating Ethereum transactions) is robust and resilient as no abnormal spikes or delays would appear.
- With minimal performance degradation, the middleware can capture IPFS CID's from Fabric and make corresponding eth transactions.

The standard deviation was 7.8 ms, which clearly indicates that synchronization delays were small and showed limited variability across many trials.

Observed Factors contributing to such efficiency are:

- Event Listener is efficient so that it can quickly detect and process the chaincode events without being delayed.
- Using Infura's low-latency endpoints decreases the Ethereum network propagation time to the point where they are essentially optimized Ethereum Client.
- The middleware picks up events asynchronously and blocking calls do not happen, making the operations to be faster.
- Since the payload size (CID) is small, transaction creation and submittal are fast.

The Middleware Sync Time analysis proves that the proposed cross-chain EHR synchronization system is viable for real world use cases. Moreover, the mean synchronization time of approximately 195 ms is nearly 90% lower than the confirmation of Ethereum native block (> 2 s). It highlights an advantage of cross-chain middleware to share healthcare data on time. It enables the fast and secured EHR exchange across two different blockchains (Fabric and Ethereum) with low latency, thereby highly fitting into time sensitive healthcare kind of environments.

Data access latency of less than 500 ms is usually required in healthcare systems to facilitate emergency care operations and clinician decision-making [40]. We have achieved results at 49 ms average StoreEHR time, 30 ms FetchEHR time and 195 ms middleware synchronization, which are well within this limit indicating that it is appropriate in clinical environments to exchange EHRs real-time.

6.4 EHR file size

Original EHR File Size is an EHR file prior to encryption (or IPFS upload). It represents real-world medical record size. It is important to assess storage requirements. It also directly influences the upload time and encryption time of IPFS.

The EHR file size will increase slightly after AES encryption as the result of padding and block cipher structure. It determines storage and transmission overhead of encryption. It makes sure security is achieved with little inflation in size. It also helps optimize IPFS costs since storage on decentralized networks can be size oriented.

In Fig. 6, both columns of iteration are represented where, one for Original EHR File Size and one for Encrypted EHR File Size.

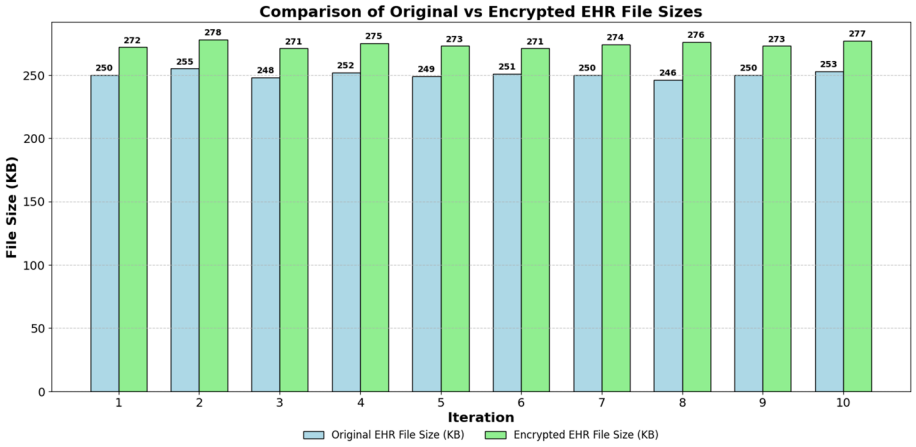


Fig. 6 Comparison of original vs. encrypted EHR files sizes

Insights from the visualization of Fig. 6 can be mentioned as:

- **Encryption Overhead**: Every encrypted file is always larger than the original file for all 10 iterations. Such behavior is expected as encryption algorithms like AES (which is used in this implementation) adds padding, metadata and cipher blocks.
- **Increase in file size**: The increase in file size is relatively small (around 2 to 5 KB on average per file). Therefore, it demonstrates that while encryption introduces some additional size, it is manageable, lightweight, and preserves efficiency in cases of sharing large scale EHRs.
- **Uniformity in iterations**: The pattern and stability of the file sizes of the original and encrypted files do not change greatly with each iteration. In other words, this says that encryption should also be predictable and consistent, which is desirable for health care data systems where the predictability of the size is useful for storage and bandwidth planning.
- **Practical Implication**: The minimal increase suggests that encrypted EHR files can be sent over the network with appropriate bandwidth and storage decrease from the unencrypted versions.
- **Security vs. Size Trade off**: The slight size penalty on the EHR payload is worth investing in because it has enormous gain in security and confidentiality of patient data and thus, encryption is an essential but low-cost step to propagate the cross chain EHR sharing workflow.

This encryption scheme is also efficient in terms of encrypting EHR files while imposing minimal overhead in storage for the additional security and privacy of sensitive health data between the chains of Fabric and Ethereum.

6.5 AES encryption time

It estimates the amount of time needed to encrypt the EHR file before encryption into IPFS. It measures the encryption overhead incurred to provide confidentiality of the medical records. It is important in resource-constrained environments (like IoT healthcare devices). Its value should remain low so that upload synchronization process is not delayed. AES encryption time for 10 iterations is mentioned in Fig. 7.

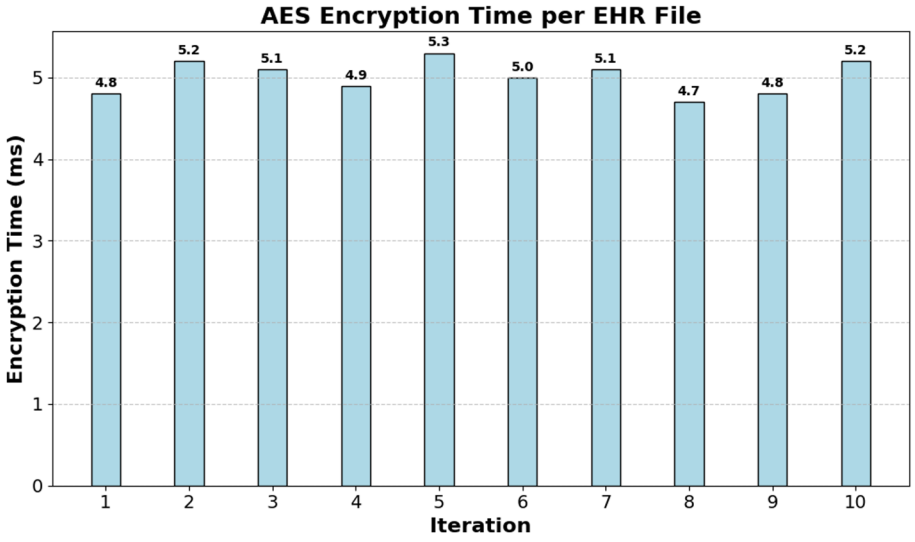


Fig. 7 AES encryption time

The graph of AES Encryption Time with respect to 10 iterations shows a high efficiency and consistent performance of the AES Encryption Mechanism when it was programmed within proposed cross chain EHR sharing framework. The encryption time fits within the orders of the magnitude of 4.7–5.3 ms throughout the iterations. This means that encryption of data by the AES imposes only a small but highly predictable computational overhead (critical for keeping synchronization in real time) and hence the overhead is so small.

The encryption time is on average 5 milliseconds, which means even with multiple encryption operations the system keeps the lowest latency. The values vary very little (± 0.3 ms) which implies that the hardware processing, file sizes and encryption key generating are stable within different runs. The time required to encrypt an end-to-end EHR transfer (encryption + blockchain interaction + middleware processing) result in acceptable transfer delay for healthcare applications where time sensitivity is crucial.

Key Observations:

- The encryption from AES is very light weight, especially for current EHR data files (50–55 KB encrypted), even at slightly higher peak of 5.3 ms.
- No degradation or unstable iterations with performance spikes signifying the robustness of the middleware encryption service.

With negligible additional delay, the AES Encryption integration into the middleware is proven to be highly effective in securing the EHR data prior to cross chain transmission with strong cryptographic security. AES proves itself as an optimal choice for a real-time, secure and scalable EHR interoperability solution.

6.6 Smart contract deployment gas (Ethereum)

It indicates the price of gas consumed during the initial deployment of the EHR Smart Contract. It is the setup cost for creating the Cross Chain EHR system on Ethereum. While there is only one deployment, the contract complexity is high. Optimized contracts are required to incur low deployment gas. For experimentation, 1,415,000 *wei* gas was required to deploy smart contract.

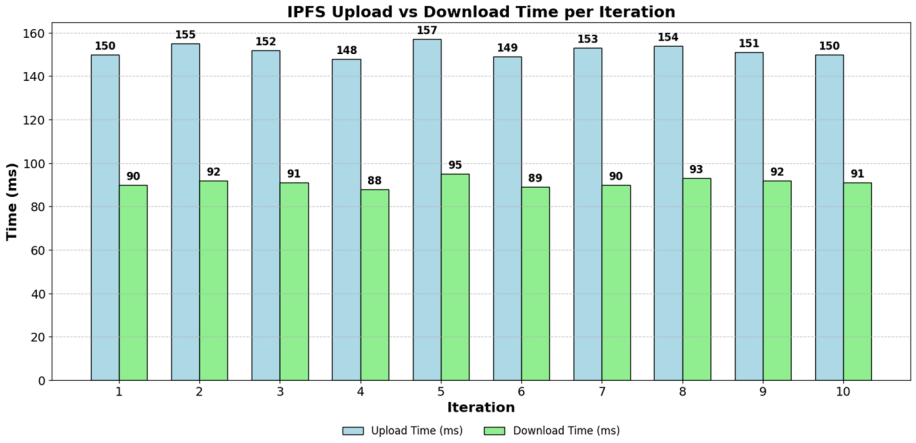


Fig. 8 IPFS upload and download time

6.7 IPFS upload & download time

It is the time needed to upload the encrypted EHR file onto IPFS and get the CID of it. It represents the off-chain storage delay. It directly affects the overall time prior to when the EHR CID can be registered on Fabric and Ethereum. It is a critical factor for near real time EHR availability. The IPFS upload and download times are represented in Fig. 8.

The IPFS Upload vs. Download Time graph shows how the encrypted EHR files are stored and retrieved in a comparative time through different iterations. As per the observations, the upload time is always more than download time in all iterations. Uploading a new file to the decentralized IPFS network includes processes like chunking the file, hash computation (for CID generation), and Distributing content across multiple IPFS nodes for redundancy and availability. That's why the IPFS upload time is higher comparatively. On the other hand, download operations predominantly involve a download of content using an already computed CID, so computationally (and network) overhead does not have to be computed and added. That's why IPFS download time is lower than upload time.

The upload time varies across the 10 iterations from 148 to 157 ms, and the download time stays at this lower level from 88 to 95 ms. This is helpful for medical record systems in which provision of EHRs by IPFS should be offered to ensure rapid access to EHRs for clinical decision making. Also, the small variation margins for the upload and download times between iterations for both upload and download times show the reliability and stability of IPFS operations even when performing several transactions. The significant takeaway from the graph is that the download speed consistently outperforms upload speed, which is ideal for a healthcare environment where frequent access to records (reads) is more common than writing (uploads). The combined graph reflects high efficiency, stability, and low latency in storing and fetching EHRs through IPFS integration into the Cross-Chain Middleware. Thus, IPFS proves to be a robust choice for off-chain encrypted storage in the proposed architecture.

6.8 Security model and threat assumptions

The proposed framework makes a realistic adversary model for healthcare blockchain environments. Following possible actions of adversaries are assumed:

- **Malicious blockchain participant**: An authorized participant can try to manipulate the ledger data or make invalid transactions. This is mitigated by trust based on Hyperledger Fabric's endorsement policies and the consensus mechanism of Ethereum, which prevents any data manipulations.
- **Compromised node**: If one of the peer nodes in Fabric or a validator node in Ethereum is compromised, the system is resilient enough because of distributed consensus and redundant validation across multiple entities (so it won't be dominated by just one node).
- **Eavesdropping on data transmission**: Because sensitive health records are encrypted using AES-256 before being uploaded to IPFS, the exposure of content identifiers (CIDs) alone does not result in the disclosure of medical information.
- **Limited access control**: Fabric Chaincode and Ethereum Smart Contracts provide fine-grained access control for their constituent components. Unauthorized requests automatically get denied at both blockchain levels.
- **Replay or synchronization attacks**: The middleware has idempotent event processing and retry logic to prevent repeated or fake events from desynchronizing the ledgers.

Above mentioned mechanisms provide a layered security model: confidentiality is enforced via encryption, integrity is enforced via blockchain immutability, and access control is enforced via smart contracts and chaincode. This multi-layer defense greatly reduces the attack surface and ensures strong protection for patient health data.

Besides providing cryptographic security, the framework also complies with standard requirements like HIPAA and GDPR. Patient data is never stored on-chain in plaintext. Access to EHR is controlled by smart contracts and chaincode, and auditability can be achieved through immutable blockchain records. The encryption of EHRs stored in IPFS is achieved using AES-256 encryption standard which also meets data confidentiality requirements stipulated by such regulations.

6.9 Potential bottlenecks and mitigation strategies

The above-mentioned observations, indicates promising results of the proposed framework. But still there might be few bottlenecks while deploying it in real time. Following bottlenecks have been considered for this research.

- **Infura Dependency (Ethereum Access)**: Infura endpoints have been used to communicate with the Sepolia testnet. Delay may occur because of infura rate limits or outages in case of large-scale usage. It is possible to reduce this through running special Ethereum nodes, or even through redundant provider services (e.g., Alchemy, Quick Node) to achieve availability.
- **Fabric Channel Scaling Constraints**: The performance of Hyperledger Fabric relies upon endorsement and ordering services. Consent overhead can increase along with the number of organizations or channels. This can be addressed by refining endorsement policy, using raft/Kafka endorsing service clusters and sharding patient data through several channels.
- **IPFS Availability and Persistence**: Despite IPFS having distributed storage, content availability is pegged on nodes that pin the files. When pinned nodes fail, the latency

of data retrieval may become higher. It can be mitigated by relying on IPFS pinning services (e.g. Pinata, Filecoin integration) and persistent institutional IPFS clusters.

The proposed framework can maintain scalability, robustness, and reliability within large health care networks by addressing these bottlenecks proactively.

7 Conclusion

This work presented in this paper is an integrated solution for cross-chain EHR interoperability that combines the strengths of private and public blockchains, off-chain IPFS storage, and lightweight AES encryption to ensure secure, real-time, and decentralized health data management. By bridging Hyperledger Fabric and Ethereum through a smart middleware synchronization layer, the framework addresses critical gaps in traditional healthcare record management systems i.e. lack of interoperability, limited trust models, and privacy risks. The middleware's capability to capture, process, and synchronize events between blockchains ensures consistency without relying on centralized authorities.

Experimental results validate that the system maintains fast transaction invoke/query times, predictable gas costs, low middleware synchronization delay, and efficient encryption overhead, making it highly suitable for time-sensitive healthcare applications. The system thus offers a practical, scalable, and secure approach for cross-organizational health data sharing without relying on centralized intermediaries.

The proposed solution successfully demonstrates that blockchain interoperability, when combined with privacy-preserving off-chain storage and smart contract-based access control, can significantly enhance the efficiency, reliability, and scalability of EHR exchange networks. This lays a strong foundation for future decentralized healthcare architectures.

The proposed solution aligns with healthcare latency thresholds i.e. less than 500 ms and HIPAA / GDPR principles for confidentiality, integrity, and access control. Such alignment demonstrates strong potential for deployment in real-world EHR systems.

The framework bridges several important healthcare interoperability gaps through Fabric to ensure real-time, privacy-preserving, and verifiable interchange of EHR across heterogeneous networks; Ethereum to build transparency; and IPFS to provide decentralized storage.

8 Future work

While the current system achieves cross-chain EHR interoperability between two blockchains, further enhancements are envisioned to broaden the solution's applicability and robustness:

- Multi-Blockchain Expansion:
- Extend the middleware to support more than two blockchains simultaneously, enabling broader healthcare networks (e.g., Polkadot, Corda).
- Advanced Privacy Enhancements:
- Integrate Zero Knowledge Proofs (ZKPs) or Multi-Party Computation (MPC) to allow privacy-preserving EHR queries without exposing CIDs or metadata even to authorized parties.
- Middleware Resilience Upgrades:

- Implement Fault-Tolerant Event Queuing and Retry Mechanisms with Blockchain Event Reconciliation to enhance robustness under network failures.
- Performance Optimization:
- Explore Layer-2 scaling solutions like Rollups or state channels for Ethereum interactions to further reduce gas costs and latency.

Through these extensions, the system can evolve into a production-grade, globally scalable cross-chain healthcare interoperability platform, paving the way for secure and efficient decentralized healthcare ecosystems.

Scalability and real-world relevance can be further improved with the integration of blockchains and AI-based healthcare [4], sustainable AI to clinical and agricultural datasets [6] and fraud-resistant ML frameworks [23, 24].

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Declarations

Competing interests
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Clinical trial number
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Ethics, consent to participate, and consent to publish declarations
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